

Roll No.....

ODD END SEMESTER EXAMINATION DECEMBER – 2025

Name of the Program: **B.Tech. (Electrical Engineering)**

Semester: 3rd

Name of the Course: **Mathematics III**

Time: 3 Hours

Course Code: TMA 301

Maximum Marks: 100

Note:

- i. All questions are compulsory.
- ii. Answer any two sub questions among a, b & c in each main question
- iii. Total marks in each main question are twenty.
- iv. Each question carries 10 marks

Q1		(20 marks)	CO1																								
(a)	Solve these questions																										
(i)	Find the probability of getting multiple of 3 in rolling a dice?																										
(ii)	Find the probability of getting 6 a sum of 6 in rolling 2 dice?																										
(b)	The joint Probability mass function is given by $p(x,y) = k(2x+3y)$, $x=0,1,2$ and $y=1,2,3$. Find (i) K. (ii) Marginal Probability distribution of X. (iii) Marginal Probability distribution of Y. (iv) Conditional Distribution of X given $Y=1$. (v) Conditional Distribution of Y given $X=1$.																										
(c)	$f(t) = \begin{cases} 2t+2 & -1 \leq t \leq 0 \\ 0 & 0 \leq t \leq 1 \end{cases}$ $f(t) = f(t+2)$ Determine the Fourier series expansion of $f(t)$.																										
Q2		(20 marks)	CO2																								
(a)	Find the probability of arranging 7 different books if 5 books are always together?																										
(b)	If the joint probability function as (x, y) is given by $f(x, y) = k : 0 \leq x < y \leq 2$. Find k and also the marginal and continuous density function?																										
(c)	Let X & Y be two R.V each taking values -1as1 having the J.P.D as given below- <table border="1"><tr><td></td><td>X</td><td>-1</td><td>0</td><td>1</td></tr><tr><td>Y</td><td></td><td></td><td></td><td></td></tr><tr><td>-1</td><td></td><td>0</td><td>0.1</td><td>0.1</td></tr><tr><td>0</td><td></td><td>0.2</td><td>0.2</td><td>0.2</td></tr><tr><td>1</td><td></td><td>0</td><td>0.1</td><td>0.1</td></tr></table> (i) Show that X & Y have different expectation. (ii) Find $E(XY)$ (iii) Find $\text{Var}(X)$ & $\text{Var}(Y)$ (iv) Given that $Y = 0$, what is the C.P.D of X				X	-1	0	1	Y					-1		0	0.1	0.1	0		0.2	0.2	0.2	1		0	0.1
	X	-1	0	1																							
Y																											
-1		0	0.1	0.1																							
0		0.2	0.2	0.2																							
1		0	0.1	0.1																							
Q3		(20 marks)																									

(a)	In a class of 75 students, there are 15 very intelligent students, 45 are mediocre and the remaining 15 are below average. The probability then an average achieving student fails 0.005, that an average achieving student fails is 0.05 and the probability of a below average achieving students failing is 0.15. If a student is known to have passed, what is the probability that he is below average achiever?	CO3
(b)	Compute the first four moments about the mean for the following frequency distribution. X 10 15 20 25 30 Y 2 5 4 9 5	
(c)	Compute the coefficient-of skewness from the following data- 25,15,23,40, 27, 25, 23, 25,20.	
Q4	(20 marks)	
(a)	Define a random variable with an example and describe its type.	CO4
(b)	The first four moments about distribution about the value 4 of the variable are 1.5, 17, -30 and 108. Discuss the Kurtosis of the distribution?	
(c)	Use Laplace transform to solve the differential equation $\frac{dy}{dx} + 2y = 10e^{3x}, x \geq 0$ Subject to the boundary condition $y = 6$ at $x=0$?	
Q5	(20 marks)	
(a)	Let X be a continuous random variable with PDF $f(x) = \begin{cases} x^2(2x + 3/2) & 0 < x \leq 1 \\ 0 & \text{otherwise} \end{cases}$ If $Y = \frac{2}{x} + 3$, find $Var(Y)$	CO5
(b)	Use Laplace transform to solve the differential equation $\frac{d^2y}{dx^2} - 4y = 24 \cos 2x, x \geq 0,$ Subject to the boundary condition $y=3, \frac{dy}{dx} = 4$ at $x=0$.	
(c)	$f(x) = x, x \in \mathbb{R}, -\pi \leq x \leq \pi.$ $f(x) = f(x + 2\pi).$ Determine the Fourier series expansion of $f(x)$.	